AI Academic Navigator

GPT-4 Powered Student Assessment in Higher Education

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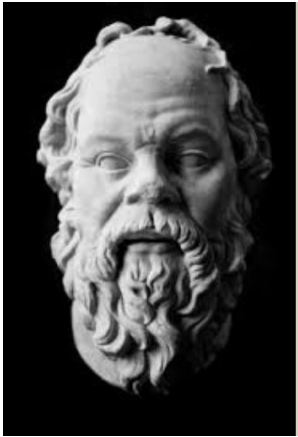
Outline

1. Motivation
2. Assessment Modules
3. System Architecture
4. Platform Demo
5. User Experience Flow
6. Future Directions



"Gnothi seauton", said Socrates: "Know thyself".

From ancient wisdom to modern AI — the mission remains the same: **help students understand themselves** to learn more effectively.

Advantages of LLM & GPT Models

1. Synthesizes across multiple results

GPT-4 can integrate data from different tests into one coherent profile.

Nori et al. (2023). *Capabilities of GPT-4 on Medical Competency Examinations.* Microsoft Research.

2. Explains complex results in plain language

AI-generated medical reports were rated more understandable by patients than doctor-written ones.

Lyu et al. (2023). *Translating Radiology Reports into Plain Language.* JAMA Network Open.

3. Gives personalized feedback to students

LLM explanations of assessment results matched expert tutor quality, improving self-regulated learning.

Kasneci et al. (2023). *ChatGPT for Good? On LLMs in Education.* Learning & Individual Differences.

Our use: All three capabilities combined in one student assessment platform.

Status Quo of Current Assessment Systems

Current Limitations

- Delayed feedback (days/weeks)
- Generic, one-size-fits-all interpretation
- Fragmented results across tests
- Low engagement ($\sim$50% completion)
- No cross-instrument synthesis

Our Vision

- **Immediate** AI-generated feedback
- **Personalized** interpretation via GPT-4
- **Integrated** multi-dimensional profiles
- **Engaging** conversational interface
- **Privacy-first** client-side architecture

Key Insight: AI augments validated instruments – it does not replace them.

Seven Validated Psychological Instruments

Module	Basis	Measures
Stroop Test	Stroop (1935)	Selective attention, cognitive flexibility
Mental Rotation	Shepard & Metzler (1971)	Spatial visualization ability
VAK Assessment	Fleming (1987)	Visual / Auditory / Kinesthetic
Grit Scale	Duckworth (2007)	Perseverance and passion
Growth Mindset	Dweck (2006)	Beliefs about intelligence
RIASEC	Holland (1959)	Career interest typology
Emotional Intelligence	Goleman (1995)	Self and social awareness

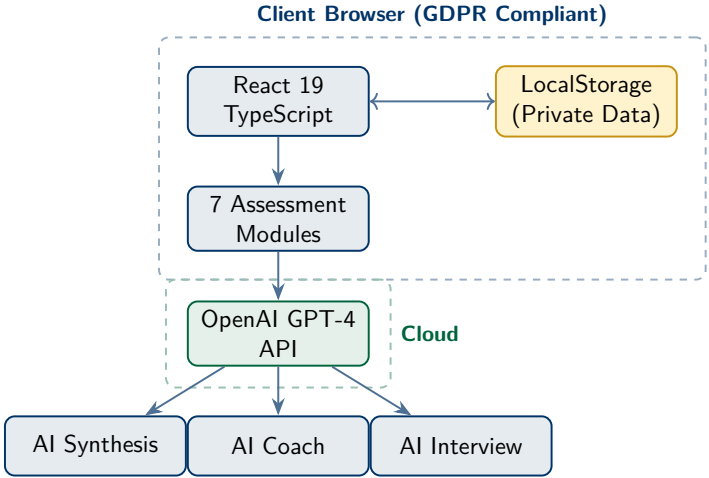
Cognitive Performance

- Stroop: 24 trials (RT + accuracy)
- Mental Rotation: 16 trials, 3 difficulty levels

Self-Report Instruments

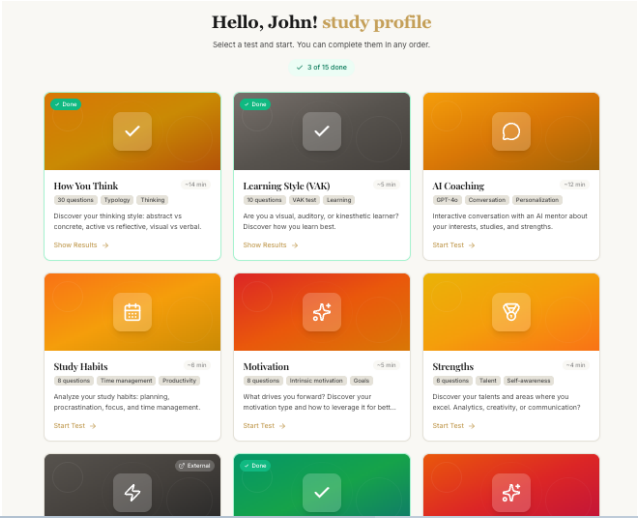
- 5 validated questionnaires
- 8–12 items each
- Likert-scale responses

Architecture: Privacy-First Design



Stack: React 19 + TypeScript · Tailwind CSS · Vite · GPT-4 API · Bilingual CZ/EN

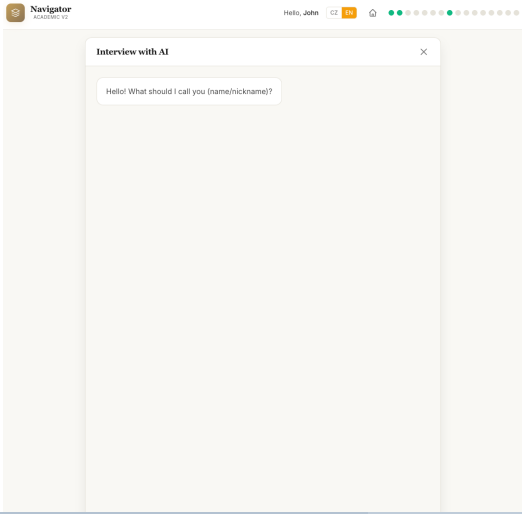
Assessment Dashboard



Features

- Gallery-style card layout
- 16 cognitive & metacognitive tests
- Visual progress indicators
- Complete in any order
- Estimated completion time per test
- Bilingual CZ/EN interface

AI Conversational Interview



GPT-4 Powered Dialogue

- Natural conversation about interests, study habits, career goals
- AI adapts questions based on responses
- Generates comprehensive "Student Passport"
- Goes beyond what structured tests capture

Key Innovation

Conversational AI explores qualitative dimensions that traditional psychometrics cannot reach.

AI Synthesis & Recommendations

John, your profile

Complete overview of all results

Results Overview | AI Synthesis & Tips | AI Coach & Plan

COMPREHENSIVE PROFILE SYNTHESIS

Your Learning Profile Summary

Based on your completed assessments, here's a comprehensive view of your learning style and cognitive profile:

Thinking Style

You show a **concrete-practical** orientation combined with an **active learning** approach. This means you prefer hands-on experiences and real-world examples over abstract theories. You learn best by doing rather than just reading or listening.

Learning Modality

Your **visual learning preference** (60%) indicates you process information most effectively through images, diagrams, charts, and visual representations. Consider using mind maps, infographics, and color-coded notes to enhance your study sessions.

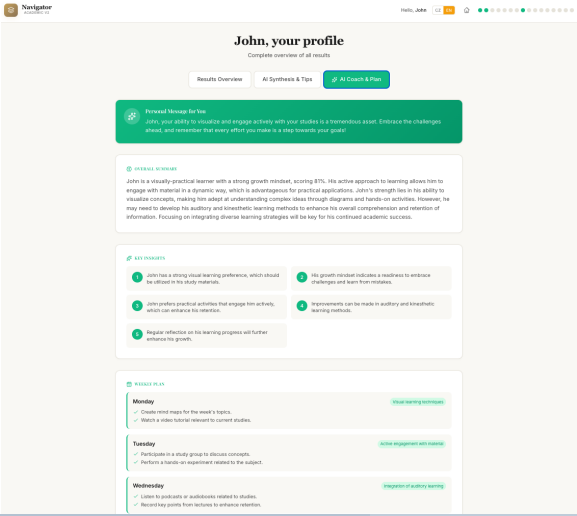
Mindset & Motivation

With a **strong growth mindset** (81%), you believe in the power of effort and practice to develop your abilities. This is a significant strength that will help you persevere through challenges and continue learning throughout life.

Key Strengths

- Visual information processing
- Active, hands-on learning approach
- Strong belief in personal development

AI Coaching Module



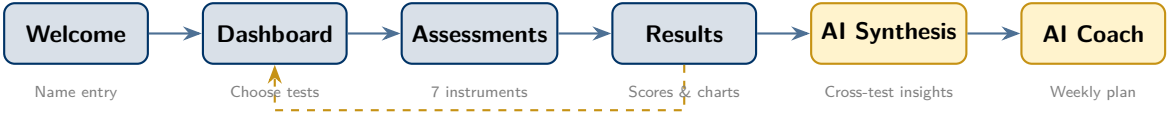
Personalized Action Plans

- Weekly study schedule tailored to cognitive profile
- Day-by-day recommendations
- Links activities to specific assessment results
- Habit tracking
- Motivational guidance

Example

Visual learner + Growth mindset → "Create mind maps for Monday's topics, watch video tutorials"

Student Journey



Assessment	AI Analysis	Action
7 validated instruments Cognitive + Metacognitive	GPT-4 cross-module synthesis Pattern identification	Personalized study plan Habit tracking & coaching

Future Directions

Next Steps

- **Longitudinal tracking** – connect assessments to academic outcomes
- **Wearable integration** – connect with measuring tools like smartwatches (HRV, stress)
- **Multi-language** – international deployment

Research Agenda

- Validation of AI recommendation quality
- Impact on student retention
- Comparison with human advisor outcomes
- Ethical framework for AI-mediated assessment

The future of AI in education lies not in replacement but in augmentation: helping students understand themselves better and chart more effective learning paths.

Thank You!

Don't hesitate to connect

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Connect with me!

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